

India COVID-19 EDA Mini-Project

Final project / submission guide based on the uploaded course materials and notebook.

1. What the final outcome should be

The final deliverable is a completed Jupyter notebook that can be opened and run in VS Code. It should show the full EDA workflow: loading the COVID-19 datasets, inspecting the data, cleaning inconsistent values, creating useful metrics, analysing trends and states, producing 10+ charts, writing observations/conclusions, and exporting the cleaned data.

This matches the course requirement for Module 1: EDA on India's COVID-19 dataset with 10+ charts and a written report.

2. Changes to make

Area	What to do
Notebook	Keep the existing notebook structure; run and complete all cells.
Environment	Use VS Code with the Python and Jupyter extensions.
Datasets	Keep the required COVID case and vaccination CSV files in the project folder. Do not rename them unless the notebook code requires it.
Cleaning	Ensure dates and numeric columns are converted correctly and inconsistent state names are standardized.
Analysis	Keep the national totals, state analysis, recovery/death rates, pivot-table analysis and vaccination analysis in the notebook.
Visualisation	Ensure at least 10 meaningful charts are produced, following the notebook's planned analysis.
Observations	Add a short, data-based observation below/after each major chart.
Conclusion	Add a final summary of the main trends discovered.
Output	Save the cleaned dataset generated by the notebook (the notebook's intended cleaned CSV output).

3. Required project flow

- Import NumPy, Pandas, Matplotlib and Seaborn.
- Load the COVID case and vaccination datasets.
- Inspect shape, data types and missing values.
- Clean date/numeric columns and standardize inconsistent state names.
- Create active cases and rate-based metrics where required.
- Aggregate national totals using groupby().
- Analyse the latest state-wise figures and identify high/low case states.
- Use pivot_table() for a summarized state/month view.
- Analyse vaccination progress.
- Create 10+ charts and interpret them.
- Write observations and a final conclusion.
- Export the cleaned COVID dataset.

4. Charts to verify

1. COVID confirmed/recovered/death trends over time
2. Daily/new case trend
3. Top states by confirmed cases

4. Recovery rate comparison
5. Top-state growth over time
6. State contribution to cases
7. Recovery-rate trend
8. Recovery rate vs death rate
9. Correlation heatmap
10. Monthly case distribution (boxplot)
11. Vaccination progress
12. Low-case state/UT comparison

5. What not to add

Do not add a machine-learning model, TensorFlow, an API, a website, A/B testing, or unrelated advanced techniques. The course materials define this project as an EDA project; the modelling stages belong to later modules.

6. Final conclusion template

“This project performed exploratory data analysis on India's COVID-19 case and vaccination data. The data was inspected and cleaned before calculating national and state-level metrics. Multiple visualizations were used to study case trends, recovery/death rates, state differences and vaccination progress. The analysis demonstrates how Pandas, NumPy, Matplotlib and Seaborn can be used to convert raw data into understandable patterns and observations.”

Replace/add details in the conclusion only when they are supported by the actual notebook outputs.

7. Submission checklist

- Notebook opens successfully in VS Code
- All cells run without unresolved errors
- Required datasets are present
- Data cleaning is completed
- Important calculations are completed
- 10+ charts are visible
- Charts have meaningful titles/labels
- Observations are written
- Final conclusion is written
- Cleaned CSV output is generated
- Notebook is saved after running it

8. Viva essentials

Question	Short answer
What is EDA?	Exploratory Data Analysis: inspecting, summarising and visualising data to find patterns.
What is Pandas?	A Python library for data manipulation and analysis.
What is NumPy?	A Python library for numerical and array operations.
What is groupby()?	Groups rows by a column or columns so calculations such as sum or mean can be performed.
What is a DataFrame?	A two-dimensional table of rows and columns in Pandas.
What is read_csv()?	It reads a CSV file into a Pandas DataFrame.
What is a pivot table?	A summarized table created by reorganizing data and applying an aggregation.
Why clean data?	To correct inconsistent, missing or incorrectly typed data before analysis.

Important: this guide describes what should be changed/verified. It does not claim to have executed the uploaded notebook or generated its actual charts, because the notebook's full executable contents and required CSV datasets were not available through the file-content index used here.